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Creating adaptive learning pathways for inquiry-based learning in school science using generative AI large language models

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ABSTRACT

This article explores the potential of generative artificial intelligence (GAI) tools to create adaptive learning pathways for inquiry-based learning (IBL) science instruction. The research compares tailored pathways from ChatGPT 4.0 and Llama 3.1 for differentiated inquiry instruction in science subjects. A qualitative approach was employed using a case study design to provide thick descriptions of Pathways from ChatGPT 4.0 and Llama3.1in supporting teachers to cater to personalised IBL instruction. Five science teachers were purposefully selected to generate and implement adaptive learning pathways for IBL instruction. Content analysis of GAI-generated pathways and teacher semi-structured interviews revealed insights into the performance of the GAI models and the usefulness of the pathways in teaching inquiry lessons. The findings revealed that LLM-based GAI engines could be used to address the individual learning needs of students in IBL science classrooms. Adaptive learning pathways showed great potential in differentiating instruction. ChatGPT 4.0 was seen to be more robust when compared with Llama 3.1. Challenges observed included difficulty keeping track of differentiated assessments, time management, ethical concerns and AI biases. These findings have implications for exploiting GAI tools when crafting IBL instruction. Some recommendations for teacher practice and research in developing adaptive learning pathways are also discussed.

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Introduction and background

From the 90s till date, Inquiry-based learning (IBL) has gained significant recognition as a practical pedagogical approach for concretising abstract scientific concepts globally (National Research Council [NRC], 2012). The IBL approach prepares students to understand the ways scientists use to explain how the natural world works. Students also become

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more scientifically literate, critical thinkers, problem-solvers and good collaborators once they understand the nature of scientific inquiry (NOSI) (Forbes et al., 2020). Unlike traditional instruction, which often emphasises rote memorisation, IBL encourages students to explore, question, and construct knowledge through active investigation (Akuma & Callaghan, 2019; Loertscher & Young, 2025). However, one serious challenge in implementing IBL in school science classrooms is the need to cater to diverse student abilities and adequate pacing. Adaptive learning pathways, powered by generative artificial intelligence (GAI), could address this challenge by providing personalised learning experiences tailored to meet the diverse needs of individual learners (Tapalova & Zhiyenbayeva, 2022).

Generative artificial intelligence (GAI) engines like ChatGPT 4.0 by OpenAI, LLaMA 3.1 by Meta, Palm 2 by Google and others use large language models (LLMs) for natural language processing (NLP), content generation, and creating interactive learning environments. The main function of GAI engines is to create new content, such as text, images, or music, by learning structures and patterns from existing data, enabling them to produce outputs that mimic human creativity (Tang, 2024; Usher & Barak, 2024).

These GAI chatbots are designed to generate human-like text, answer questions, and provide explanations based on vast training data and 'web crawling' (Cooper, 2023; Zhai, 2023). Recent advancements in AI and machine learning have enabled LLMs not only to understand and generate text but also to engage in complex tasks such as tutoring, curriculum design, and providing adaptive feedback (Brown, 2020; Radford et al., 2019). In different educational settings, LLM-based GAI tools have been explored for various applications, including creating learning content, automated grading and personalised tutoring (Gupta et al., 2024; Zhai & Nehm, 2023). However, their potential to support IBL by creating adaptive learning pathways catering to students' needs has not been fully explored to the best of our knowledge.

Adaptive learning pathways are customised learning experiences based on individual student data, such as prior knowledge, learning style, and performance indicators (Gli-gorea et al., 2023), which are created to cater for specific student needs. Teachers can leverage GAI to design dynamic learning environments that respond to the individual needs of students as it pertains to inquiry-based learning (IBL) instruction. The current study investigates how GAI tools can create adaptive pathways for IBL in school science subjects with a particular focus on Physics, Biology and Chemistry. Some of the study's contributions include an explication of how GAI tools can be prompted to provide pedagogical assistance to teachers for creating differentiated IBL instruction. Insights are also provided on the multi-theoretical framework that underpins GAI-driven differentiated instruction for IBL. The aim, objectives and research questions captured below provide insights into the inquiry.

Research aim

The main aim of this research is to investigate how LLM-based GAI tools can be used to create adaptive learning pathways that support IBL instruction in school science. To attain this aim, the following objectives were set:

- To evaluate standard prompts for IBL in Biology, Chemistry and Physics, which show student differentiation.

- To establish the mechanisms by which LLM-based GAI tools generate personalised IBL pathways for students from standard prompts.
- To evaluate teachers' views on the effectiveness of the AI-generated adaptive pathways for pedagogical purposes in IBL science classrooms.
- To identify the limitations and challenges of using different LLM-based GAI tools in creating adaptive pathways for IBL.

The research questions posed for this inquiry include:

1. How do GAI LLMs generate differentiated learning pathways for IBL from different standard prompts?
2. What are science teachers' views of the effectiveness of GAI LLMs in creating adaptive pathways for IBL pedagogy in school science subjects?
3. In which ways are the different LLM-based GAI tools limited when generating personalised adaptive learning pathways for IBL?

Theoretical lens

The study is underpinned by an intersection of the theory of differentiated instruction and Vygotsky's zone of proximal development (ZPD). The theory of differentiated instruction, as suggested by Tomlinson (1999), emphasises that the approach of 'One-size-fits-all instruction' is not a practical pedagogical approach due to the diversity that is usually seen in student background knowledge and abilities, irrespective of students belonging to the same age bracket (Tomlinson, 1999, 2001). As students differ widely, there seems to be no fixed teaching and learning methods that can address all students' learning needs. Progress rates may differ in the same group because of differences in student cognitive development, instructional strategies, and teaching materials. This then poses some challenges when teachers plan for instruction.

Vygotsky's ZPD exploits the work of mediation that can be done between what students can do independently without assistance and what they can only accomplish with guidance (Vygotsky, 1987). The ZPD highlights how crucial it is to provide support for challenges that are beyond a student's current capabilities using different mediators (Smagorinsky, 2018; Yeoh et al., 2025). The interesting twist in this theory is that what a student (A) may need in the ZPD may not be the same as what another student (B) needs. Hence, differentiated instruction is crucial even within the ZPD. Figure 1 below shows the intersection between the theory of differentiated instruction and the ZPD.

Figure 1 describes the intersection between the theory of differentiated instruction and the Zone of Proximal Development (ZPD). From Figure 1, one can see clearly that within the ZPD, there is a need for differentiated mediation. Differentiated mediation caters for the targeted support of students with assistance tailored to meet students' individual needs, as well as providing personalised scaffolds that would allow the student to attain higher levels of knowledge (Al-Hamdani & Yousif, 2025; Wen & Cai, 2024).

This multi-theoretical approach was best suited for this investigation as it anchored the nuances under which personalised learning pathways facilitated by GAI tools could be created for inquiry instruction. GAI chatbots could adjust IBL task complexity

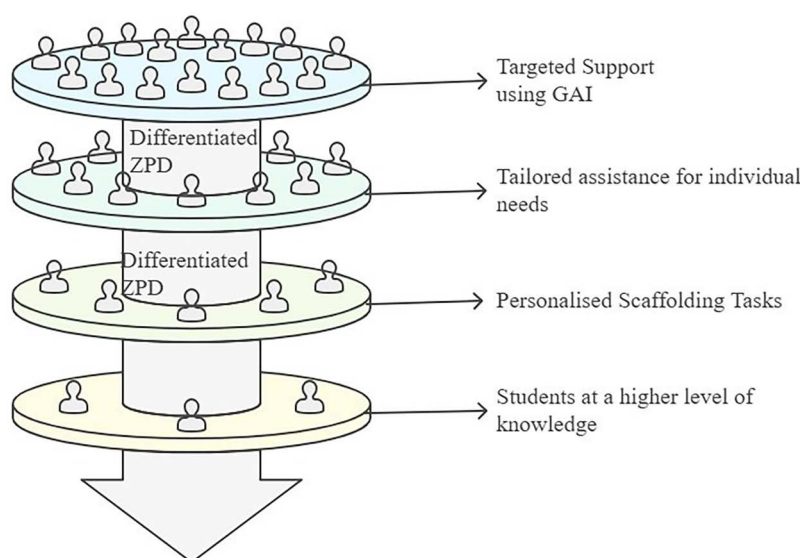


Figure 1. The intersection between differentiated instruction and the ZPD.

to align with individual students' ZPD (Al-Hamdani & Yousif, 2025). This kind of differentiation ensures that students are not too bored with the simplicity of a scaffolded IBL task nor overwhelmed by tasks that are too difficult, based on their cognitive abilities (Yeoh et al., 2025).

Challenges in inquiry-based learning (IBL) instruction

There are several challenges with inquiry instruction at the level of school science subjects. Some of these challenges include limited teacher preparedness, resource constraints, and classroom management difficulties (Author, 2016). More times than not, teachers are inadequately prepared for the implementation of IBL pedagogy. This impedes the way facilitation and assessments take place in IBL settings (Lazonder & Harmsen, 2016; Lederman et al., 2021). Other deficiencies lie in resource limitations, including laboratory access and a shortage or lack of materials and technological tools, which hinder hands-on experimentation essential to IBL (Akuma & Callaghan, 2019; Gholam, 2019). Also, managing many student groups in an inquiry situation and ensuring that engagement is maintained can also be very challenging for teachers, considering that students may require more support when navigating open-ended investigations (Gholam, 2019; Mesci et al., 2020). These issues collectively make IBL instruction quite challenging and require teachers to exploit several tools that could enhance the process.

Relating IBL challenges to individual student needs

One of the primary challenges in inquiry-based learning (IBL) is the difficulty of addressing individual student needs through differentiated instruction. IBL requires students to

engage in open-ended questioning, solve problems, and think critically, which can be challenging for students with varying prior knowledge, skills, and learning styles. A lack of differentiation can hinder the effectiveness of IBL, as students who are either advanced or struggling may not receive the appropriate level of support needed for their varying levels of comprehension (Hmelo-Silver et al., 2007).

What becomes the trend is that without differentiated learning, students of advanced cognitive abilities may become disengaged, finding IBL tasks to be insufficiently challenging, while students with other cognitive abilities or skills may struggle with overly complex IBL tasks (Tomlinson, 2001). Furthermore, some research findings show that a one-size-fits-all approach in IBL instruction creates disparities in engagement and learning outcomes, with some students being unable to keep up with the demands of independent inquiry (Mesci et al., 2020; Minner et al., 2010). With the awareness of this difficulty, differentiated IBL instruction and differentiated scaffolding, where students receive support tailored to their individual needs, whether in the form of guiding questions, structured feedback, or alternative resources that align with their abilities and interests, could be the way forward for twenty-first-century instructors. Without these supports, students risk becoming either frustrated or disengaged, ultimately limiting the benefits of the inquiry process in diverse classrooms (Darling-Hammond et al., 2008).

Personalised learning pathways

Personalised learning pathways are an approach to teaching that tailors learning experiences to suit the individualised needs, learning styles and abilities of each student (Chen et al., 2020; Walkington & Bernacki, 2020). Instead of a standardised curriculum for all, Personalised learning pathways allow students to capitalise on their preferred learning styles, pace learning to suit learning abilities, the depth of their prior knowledge and other variables like the availability of learning resources.

Some of the main benefits of creating personalised learning pathways are that they are student-centred, cater to pacing, give access to a variety of teaching methods that address the learning objectives, and are competency-based, allowing students to be assessed based on their mastery of specific skills and knowledge, rather than on time spent in class (Al-Hamdani & Yousif, 2025; Bernacki et al., 2021). Crafting personalised learning pathways entails that learning objectives, methods of instruction, and pace be varied, which is mainly supported by the theory of differentiated instruction. Other variables like cognitive abilities, preferred learning styles and preferences, and prior knowledge always come to the fore when planning for differentiated instruction (Walkington & Bernacki, 2020). The ultimate gain of Personalised instruction is that students may perform better both academically and socially (Yang & Weng, 2023).

Despite the benefits of creating personalised pathways for students, several challenges could limit the adoption of this method of instruction, including time constraints, a lack of sufficient resources to differentiate instruction, complexities in developing assessments that cater to the differentiated pathways, inadequate skills for differentiating instruction as well as threats to group interactions and collaborative learning. Notwithstanding, these challenges can be addressed by ensuring that the right balance is struck between what students do in the individual pursuit of knowledge and how collaboration with their peers is integrated into any learning scenario. GAI

tools can also be integrated to mitigate some time constraints associated with developing adaptive learning pathways.

With the possibilities that lie within personalised instruction, the reality in schools and higher institutions of learning in developing countries is different in that instructional practices are mostly standardised with little or no flexibility to accommodate the individual needs of students (Rojas & Chiappe, 2024). Some of the reasons why this is the case include the fact that a lot has to go into pre-examination of student characteristics from different learning analytics data and context data to create a successful learning pathway for individual students (Tang et al., 2019). Generative AI and other technologies can be used to alleviate some of the burdens that are placed on instructors in this regard.

Large language model-based generative artificial intelligence (GAI)

Large Language Models (LLMs) are advanced models of artificial intelligence trained to handle enormous amounts of textual data used for understanding, generating, and responding to human language. Being hosted in deep learning architecture, LLMs, especially transformer-based models like GPT (Generative Pre-trained Transformer), can process and produce coherent and contextually relevant text in different domains (Ferrara, 2024; Yeh, 2025). Generative Artificial Intelligence (GAI) refers to a subset of artificial intelligence systems designed to produce new, original content by learning from existing data. These systems are usually propelled by machine learning models, including LLMs, that aid them in the generation of text, images, videos and even code.

When OpenAI released its LLM-based generative artificial intelligence (AI) chatbot ChatGPT in late 2022, a different era of technological evolution became imminent in all domains of life. The Chatbot offered engaging text-based conversations exhibiting intelligence and a humanlike tone. From then till now, GAI chatbots have continued to evolve and intersect with all disciplines of life. In education, GAI chatbots are actively being exploited to understand their impact on educational discourses based on their ability to change learning environments (Chen et al., 2020; Zhai & Nehm, 2023; Zhang et al., 2023). Large language model-based GAI bots have been able to modify instructional processes, designs and assessments (Chen et al., 2020; Xu & Ouyang, 2022). With the impact these AI tools have in education, more solutions and support for teaching and learning are being realised, whereby large numbers of students are enjoying Personalised tutoring using tools like *Khanmigo* and other AI tutors; teachers can create and assess tasks faster than ever before and explore real-time analysis of student performance. These numerous potentials of GAI tools have long-term implications for education (Yang, 2022) and raise concerns about the impact of AI-generated data on standard educational assets like textbooks and teachers.

Research method

This study adopted an exploratory case study design (Yin, 2018) to create and study adaptive learning pathways using two LLMs, GPT 4.0 and Llama 3.1. This design was preferred as it provided flexibility for an open-ended inquiry of the GAI tools and the freedom to provide rich contextual data from content analysis, GAI engines, and

teachers' lived experiences (Yin, 2018). The selected cases of GAI tools were based on an initial open-ended survey of 485 mathematics and science teachers in schools across all the provinces in South Africa. The initial survey is an open-ended survey which has five key questions created by the researchers with the aim of establishing teachers' use of Gen AI tools, the specific tools they use, the price point of the tools (free or paid), why they prefer the tools over others, and the purpose for which the tools are used. This was given to the teachers in a national cluster where the first author facilitates and professionally develops science teachers as a baseline assessment of teachers' Gen AI use in practice. From this demography, 82.47% ($n = 400$) of the teachers indicated that ChatGPT 4.0 was their preferred AI tool for every pedagogical idea. In comparison, 12.37% ($n = 60$) indicated that they use Llama 3.1 on the go, and the remaining 5.15% ($n = 25$) mentioned other GAI tools like Co-pilot by Microsoft and Gemini by Google for supporting their practice.

Based on this distribution of the preferred AI tools, GPT4.0 and Llama 3.1 were considered the best GAI tools for this case study because the five participant science teachers that the researchers sourced for the in-depth study predominantly use these two GAI chatbots. The teachers, P1 and P5 (Pseudonyms), who predominantly used Llama 3.1, indicated that Llama 3.1 was suitable for their use as it was integrated with WhatsApp and Facebook and could draw imaginary images that they used in the inquiry-based teaching practices, depending on the prompt they provided.

Data collection and analysis

Participants

Five ($n = 5$, three females and two males) science teachers who had taught Biology, Chemistry and Physics with teaching experience ranging between 3–7 years were conveniently sampled to participate in this study. Although the population from which the participants were selected consisted of approximately 485 teachers, a purposive sample of five participants was selected to ensure consistency with the case study approach, which prioritises depth over breadth (Yin, 2018). The five participants allowed for intensive exploration of teachers' experiences, classroom practices, and perspectives through multiple data sources, ensuring data saturation while maintaining rigour. The participants were chosen because they represent information-rich cases aligned with the objectives of the study, they work as part of schools of specialisation in science and technology, and above all they were willing to try GAI tools in their teaching.

Table 1. Subject, Grade, and GAI use details.

Pseudonym	Subject/grade taught	Years of experience	Preferred GAI
P1	Life Sciences Grade 10 and 12	3	Llama 3.5 most times and ChatGPT 4.0
P2	Physical Sciences Grade 10 and 11	4	Chat GPT 4.0. new to using Llama 3.1
P3	Physical Sciences Grade 10 and 12	6	ChatGPT 4.0 never used Llama
P4	Physical and Life Sciences Grade 10,11 and 12	7	ChatGPT 4.0 mainly
P5	Physical Sciences Grade 11 and 12	7	Llama 3.5 almost all the time, and some ChatGPT 4.0

Table 1 below summarises the grades, subjects, preferred AI tool and years of teaching experience that the selected teachers possess.

From **Table 1**, the teachers' details regarding the subjects, grades and their preferred GAI tools are highlighted. The selected teachers also work closely with the researchers in a technology integration drive within special science and technology schools identified by the Department of Basic Education (DBE) in South Africa. These schools support the use of diverse educational technologies for teaching and learning. All participant teachers have experience with using simulations, virtual reality, augmented reality and diverse virtual laboratories for their pedagogical practices. They all have between 3–7 years of experience teaching either Biology (Life Sciences) or Physics and Chemistry (Physical Sciences).

Generative AI tools

For this study, ChatGPT 4.0, developed by OpenAI, which is an open-source successor to earlier GPT models like ChatGPT 3.5 and Llama 3.1, Meta's latest GAI tool in the Large Language Model for AI Assistance (LLaMA) series, was preferred from survey insights. GPT4.0 is built to understand and generate human-like text through deep learning and can be optimised to cater for several other analytical modalities (Cooper, 2023; OpenAI, 2024). Released in 2024, ChatGPT 4.0 supports fine-tuning and plugin integration, increasing its versatility for various applications. Llama 3.1, on the other hand, deploys enterprise solutions and efficient training (Meta AI, 2024). The bot's key affordance is low-resource accessibility while maintaining excellent language capabilities. Both ChatGPT 4.0 and Llama 3.1 represent GAI tools that can deliver in task-oriented scenarios, and which the five teachers used regularly.

Procedure

In developing personalised learning pathways that science teachers could use for IBL instruction, some differentiation criteria were considered, including subjects, resource availability for IBL, students' prior knowledge, students' cognitive abilities and students' remediation needs. The data collection process was three-fold:

Firstly, based on the participants' preferred AI tool, they were asked to choose a topic from any of the science subjects they teach, develop an IBL-integrated lesson plan, which considered the context, lesson objectives, the strengths and weaknesses of learners, as well as the exact IBL tasks that they intended for the AI tool to personalise and create adaptive learning pathways.

Secondly, the teachers were then asked to engage in prompting of both GAI engines based on the first steps taken in selecting the topic, grade, context, etc. Teachers obtained all the data (AI-generated outputs) from focused prompting based on the subjects, lesson plan and learner differentiation criteria (student performance, prior knowledge levels, suggested activities, scaffolding strategies and remediation suggestions) that they identified within their own classroom context. The output data collected by each teacher from the two GAI engines on their planned IBL lesson was then analysed for patterns and variations through content analysis using a coding rubric/checklist created by the researchers. For example, **Table 2** shows an excerpt from the checklist that was used to analyse the content of the GAI tools' ability to create adaptive pathways based on differences in

Table 2. Checklist for content analysis of GAI tools.

Differentiation Criteria	Evaluation Aspects	GAI chatbot performance	
		Basic (1)	Advanced (2)
Low Performers	Identifies low performers and tailors content to their needs.	Assess if the tool identifies low performers via quiz scores and provides simplified Biology, Chemistry, or Physics content.	Evaluates if the tool uses self-assessments to identify low performers and offers basic modules (e.g. introductory Chemistry concepts).
	Provides scaffolded support to enhance understanding.	Checks if the tool offers step-by-step explanations (e.g. basic cell structure in Biology).	Check if the tool provides remedial tutorials (e.g. fundamental Physics equations).
	Tracks progress and adjusts tasks for low performers.	Evaluates if the tool updates tasks based on real-time performance in classroom quizzes.	Assess if the tool modifies module sequences for low performers in self-paced learning.
Middle Performers	Identifies middle performers and provides appropriately challenging content.	Assess if the tool recognises middle performers and offers balanced tasks (e.g. stoichiometry in Chemistry).	Evaluates if the tool assigns intermediate modules (e.g. moderate Physics problems).
	Balances scaffolding and independent tasks for engagement.	Checks if the tool provides guided inquiry tasks (e.g. hypothesis testing in Biology).	Verifies if the tool offers self-directed tasks with moderate complexity (e.g. chemical equilibrium).
	Tracks progress and adjusts pathways for middle performers.	Evaluates if the tool adapts tasks based on classroom performance trends.	Assess if the tool adjusts module difficulty based on ongoing online performance.
High Performers	Identifies high performers and provides advanced content.	Assess if the tool identifies high performers and offers enrichment (e.g. advanced Physics problems).	Evaluates if the tool assigns advanced topics (e.g. quantum Chemistry for online learners).
	Encourages independent exploration and critical thinking.	Checks if the tool suggests open-ended projects (e.g. designing a Biology experiment).	Verifies if the tool provides research-based tasks (e.g. Physics simulations).
	Tracks progress and adjusts challenges for high performers.	Evaluates if the tool updates tasks to maintain engagement in classroom settings.	Assess if the tool offers specialised modules for high performers in self-paced learning.
Differentiation Criteria	Customises content based on learning pace.	Checks if the tool offers adjustable pacing for group vs. individual tasks in a classroom.	Verifies if the tool allows self-paced progression with deadlines or adaptive pacing suggestions.
	Adjusts content complexity based on learner needs.	Assess if the tool provides tiered explanations (e.g. basic vs. detailed chemical bonding).	Evaluates if the tool offers advanced readings or simplified summaries based on learner preferences.
	Incorporates diverse learning modalities (e.g. visual, textual, interactive).	Checks if the tool suggests videos, diagrams, or text-based explanations for diverse learners.	Verifies if the tool integrates interactive simulations or text-based modules for online learners.

(Continued)

Table 2. Continued.

Differentiation Criteria	Evaluation Aspects	GAI chatbot performance	
		Basic (1)	Advanced (2)
Suggested Activities	Proposes inquiry-based activities tailored to learner needs.	Assess if the tool generates subject-specific tasks (e.g. Biology experiment on osmosis).	Evaluates if the tool suggests research projects or simulations (e.g. Physics projectile motion).
	Recommends collaborative vs. individual tasks based on context. Aligns activities with curriculum standards and IBL principles.	Checks if the tool proposes group discussions or peer reviews for classroom learners. Assesses alignment with high school standards (e.g. NGSS for science inquiry).	Verifies if the tool suggests self-directed projects or forums for online learners. Evaluates alignment with undergraduate learning outcomes (e.g. critical analysis in Chemistry).
Differences in Prior Knowledge	Assesses and accounts for prior knowledge levels.	Checks if the tool evaluates foundational knowledge via pre-tests (e.g. basic Physics concepts).	Verifies if the tool uses diagnostic questions to gauge prior coursework (e.g. organic Chemistry).
	Adapts content to bridge knowledge gaps.	Assess if the tool provides remedial content for gaps in Biology (e.g. cell structure basics).	Evaluates if the tool offers review modules for gaps in Physics (e.g. kinematics fundamentals).
	Integrates advanced content for learners with strong prior knowledge.	Checks if the tool offers enrichment tasks (e.g. advanced stoichiometry in Chemistry).	Verifies if the tool provides specialised topics (e.g. quantum mechanics for advanced learners).

student performance, differentiation criteria, suggested activities, scaffolding strategies and remediation suggestions.

Table 2 provides an excerpt of the coding rubric for assessing how each of the GAI tools performed when differentiating instruction for different learning pathways as presented by the participant teachers. The GAI chatbot suggestions were classified as either basic (1) or advanced (2) based on the level of differentiation that was suggested for each differentiation criterion provided by teachers in their prompts.

Thirdly, participant teachers used the adaptive learning pathways generated by GAI tools to implement the core ideas in their teaching using IBL. The researchers observed five post-AI engagement lessons to collect data on the implementation of the adaptations provided by the GAI tools and further triangulated the data by conducting five individual semi-structured interviews with the participant teachers.

The aim of the semi-structured interviews was to get the insights of participants' implementation experiences and to craft future directions on how GAI could be integrated into IBL instruction in a manner that caters to all learners. Data gathered through interviews was transcribed verbatim and analysed thematically to obtain insights that relate to the pedagogical value of GAI tools in creating personalised learning pathways for science students engaging in IBL tasks. A detailed codebook was developed to classify the different codes and categories which emerged from interview data for commonality and patterns. Six questions, directly related to teachers' experiences of using GAI for differentiated IBL instruction, were posed (questions added to Appendix A).

All the relevant permissions were obtained from the participants and the institutions where the research was conducted. Measures of credibility and trustworthiness were ensured by employing participant verification of transcribed text to make sure there were no misquotations or misrepresentations.

Interrater reliability for excerpts generated from the GAI pathways was calculated using independent scores assigned by two coders, using two categories (basic and advanced), using Cohen's Kappa (Cohen, 1960). The observed proportional agreement (Po) was found to be .90, while the probability of agreement expected by chance across the two categories (Pe) was .25. Applying the formula $\kappa = (Po - Pe) / (1 - Pe)$, Cohen's Kappa was calculated to be approximately .87. This value indicates a very good agreement between the two coders (Landis & Koch, 1977) in their assessment of GAI pathways. The same procedure was used to validate interview and observation data sets.

Results

This section presents the results in a way that will answer the research questions posed.

GAI's differentiate learning pathways for IBL tasks

Participant teachers followed the directions given by the researchers to prompt the GAI chatbots with six standard prompts each for differentiated IBL instructional pathways in their respective science subjects. The outlook for the standard prompts can be seen in Figure 2 below.

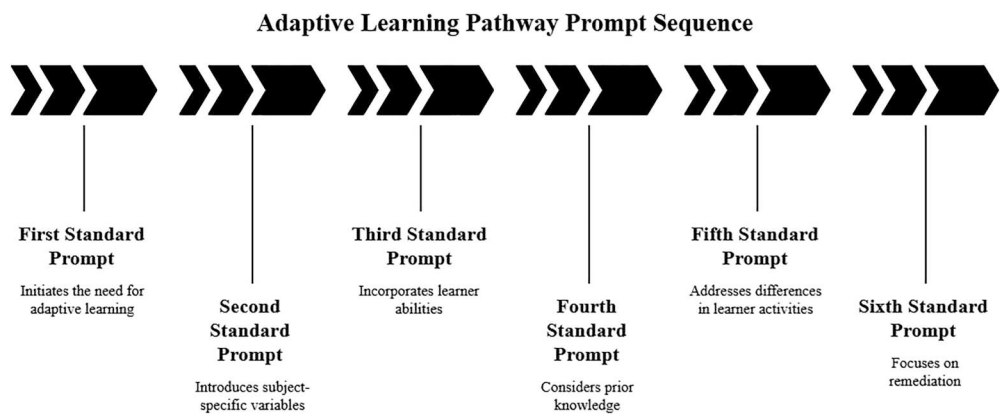


Figure 2. Adaptive learning pathway prompts.

Figure 2 shows the path that the teachers followed to arrive at the six standard prompts that addressed the different topics and subjects participants planned to teach, and the different differentiation criteria. The six standard prompts included: 1. A prompt to let the GAI tool know you wish to adapt instruction for different learners, 2. Prompt 2 deals with subject-specific variables, 3. Prompt 3 looks at varying abilities of learners, 4. Prompt 4 considers prior knowledge, prompt 5 highlights the need for different activities, and 6. Prompt 6 focuses on remediation tasks for different learners. For example, P1, who teaches Life Sciences in Grade 10, asked, ‘Hi bot, create a personalised learning Pathway for inquiry-based learning for 25 Science learners who take the subject of Life Sciences? I wish to use an IBL approach to teach them the factors that affect the rate of photosynthesis.’ The lesson she planned was based on the topic of ‘photosynthesis’, and the outcomes of the lesson were for learners to investigate the different factors that affect the rate of photosynthesis and be able to report back on their findings.

The sample prompt above was posed to both ChatGPT 4.0 and Llama 3.1, and the two outputs from the GAI were obtained and captured as the prompting sequence continued with different separation criteria based on the nature of learners in her class. Table 3 below shows a sample of the excerpt generated from the teacher P1’s first standard prompt from both GAI engines. Table 3 below shows excerpts from the two GAI models.

Table 3 presents an excerpt from the first level of prompts posed by teacher P1 and the GAI-crafted ideas for adaptive, personalised learning pathways for IBL with some levels of differentiation. These pathways were coded and categorised to show commonalities and differences. From Table 3, one can see that with detailed prompting, based on the parameters that a teacher or instructor wishes to adapt learning pathways, ChatGPT 4.0 and Llama 3.1 could tailor specific scenarios that a practising teacher can implement, looking at the different learning needs in their classroom. The generated adaptive learning pathways cater for the different needs of students in terms of available resources for IBL, varying levels of prior knowledge, preferred learning styles, previous performance in IBL tasks and cognitive abilities.

Findings from all generated data show that (GAI LLMs) like ChatGPT and LLaMA support teachers to a greater extent in creating adaptive learning pathways for inquiry-based learning (IBL) in science subjects. Both engines were able to personalise

Table 3. Sample subject-specific prompt to create adaptive learning pathways.

Qualifier/Prompt		ChatGPT 4.0			
Create a personalised learning Pathway for inquiry-based learning for 25 Science learners who take the subject of life sciences? I wish to use an IBL approach to teach them the factors that affect the rate of photosynthesis.	'Here's a personalized learning pathway for 25 Life Sciences learners using an Inquiry-Based Learning (IBL) approach to explore factors that affect the rate of photosynthesis. This pathway integrates scaffolding, learner diversity, inquiry stages, and assessment, and is ideal for settings with or without laboratory access. Personalized IBL Learning Pathway: Factors Affecting Photosynthesis Learning Goal  Students will explore and understand how light intensity, carbon dioxide concentration, temperature, and other factors influence the rate of photosynthesis. Stage 1: Engage (Personalized Exploration)  Objective: Activate prior knowledge and spark curiosity. Activities - Watch a video: 'How Plants Make Their Food' (EdPuzzle or YouTube) - Prompt journal: 'Why do you think some plants grow faster than others?' - Pose a question: 'What might affect the rate at which photosynthesis happens?' Stage 2: Explore (Hands-on or Simulated Inquiry)  Objective: Investigate key variables through experimentation. Activities - In small groups, learners plan and conduct experiments (real or virtual) to test one factor (e.g. light, CO₂, temperature). - Use digital simulations like PhET Photosynthesis Lab if no lab is available. - Record data in observation tables. Stage 3: Explain (Knowledge Construction)  Objective: Build scientific understanding using evidence collected. Activities - Present findings through posters, videos, or verbal reports. - Teacher facilitates a concept mapping activity to link variables to the rate of photosynthesis. - Introduce scientific vocabulary and concepts. Stage 4: Elaborate (Transfer and Extension)  Objective: Apply learning to new situations.				
			Grouping Individually	Differentiation Provide sentence starters for those with literacy barriers	Tools/Resources - Visual aids- Translated prompts- EdPuzzle (with captions)
			Grouping Small groups of 3-5	Differentiation Assign roles (scribe, leader, speaker). Use templates for experimental design.	Tools/Resources - Virtual lab- Charts, timers, thermometers- PhET Simulation (https://phet.colorado.edu)
			Grouping Whole class	Differentiation Concept maps with hints for struggling learners . Use audio explanations for low literacy students .	Tools/Resources - Google Slides- Canva- Anchor charts

(Continued)

Table 3. Continued.

ChatGPT 4.0			
Activities - Learners choose one of the following: (a) Investigate photosynthesis in aquatic plants (b) Explore how farmers optimise photosynthesis in greenhouses (c) Interview a local gardener on plant growth challenges	Grouping Individually or pairs	Differentiation Give learners choice to pick a context. Scaffold interviews with question templates.	Differentiation - Internet access- Local visits- Infographic tools
Stage 5: Evaluate (Reflection and Assessment)			
Objective: Assess inquiry process, conceptual understanding, and application.			
Activities - Learners complete a reflection: 'How did our experiment help you understand photosynthesis?' - Short quiz (adapted for various levels) - Rubric-based assessment of project or presentation	Grouping Individual	Differentiation Multiple formats: written, oral, drawing - Use self-assessment checklists	Tools/Resources - Google Forms- Teacher-created rubric- Reflection prompts
Personalisation Features			
• Flexible grouping: High-achieving learners support others. • Multimodal content: Video, text, simulations, oral explanations. • Choice: Learners select experiments, presentation formats, and elaboration tasks. • Feedback: Ongoing peer and teacher feedback during each phase.			
Assessment Aligned to Inquiry and CAPS			
Type Formative	Tool Exit slips, checklists		Focus Understanding of variables and experimental design
Summative	Mini-project, quiz		Application of inquiry skills and photosynthesis content
Reflective	Journals, self-assessment		Metacognition and conceptual clarity
Llama 3.1			
' , Here's a personalized learning pathway for 25 Science learners using an Inquiry-Based Learning (IBL) approach to explore factors affecting the rate of photosynthesis:			
Learning Objectives:			
Understand the process of photosynthesis and its importance			
Identify and investigate factors affecting the rate of photosynthesis			
Analyze and interpret data to draw conclusions			
Develop critical thinking, problem-solving, and collaboration skills			

(Continued)

Table 3. Continued.

Qualifier/Prompt	ChatGPT 4.0
	Learning Pathway: Phase 1: Introduction and Exploration (2-3 class periods) Introduction: Introduce the concept of photosynthesis, its importance, and the factors that affect its rate. Use multimedia resources, diagrams, and real-life examples to spark curiosity. Guided Inquiry: Ask learners to brainstorm and list potential factors that might affect the rate of photosynthesis. Encourage them to share their ideas and discuss as a class. Grouping and Roles: Divide learners into groups of 4–5 and assign roles: Researcher, Experimenter Data Analyst, Recorder Presenter Phase 2: Investigation and Data Collection (4-5 class periods) Research Question: Each group chooses a factor to investigate (e.g. light intensity, temperature, CO2 concentration, water availability). Experimental Design: Learners design and conduct experiments to test their research question. Provide guidance on experimental design, data collection, and safety protocols. Data Collection: Learners collect and record data, using tools like graphs, tables, or charts. Phase 3: Analysis and Conclusion (2-3 class periods) Data Analysis: Learners analyse and interpret their data, using statistical methods and graphical representations. Conclusion: Groups draw conclusions based on their findings, relating them to the research question. Peer Review: Groups present their findings to the class, receiving feedback and suggestions for improvement. Phase 4: Reflection and Application (1-2 class periods) Reflection: Learners reflect on their learning experience, discussing challenges, successes, and areas for improvement. Application: Learners apply their understanding of photosynthesis and its factors to real-world scenarios, such as: Optimizing crop growth in agriculture Understanding the impact of climate change on plant productivity Assessment: Formative assessments during each phase to monitor progress and provide feedback Summative assessment of the final presentation and written report Differentiation: Provide varying levels of complexity in research questions and experimental design Offer additional support or challenges for learners who need it Encourage learners to take ownership of their learning pathway Resources: Laboratory equipment (e.g. lamps, thermometers, CO2 sensors) Data analysis software or graphing tools Online resources (e.g. simulations, videos, interactive diagrams) Peer-reviewed articles or textbooks for background information Tips and Variations: Consider using a flipped classroom approach, where learners watch videos or complete readings at home before class Incorporate technology, such as sensors or data loggers, to enhance data collection and analysis Encourage learners to design and propose their own experiments, promoting autonomy and creativity By following this learning pathway, your learners will engage in authentic scientific inquiry, developing essential skills in critical thinking, problem-solving, and collaboration while exploring the fascinating world of photosynthesis!

*** Most text in Table 3 is AI-generated with slight edits to humanise

instruction based on learner needs, prior knowledge, and level of performance. The models analyse teachers' input on the differentiation criteria to generate tailored scaffolding, suggest differentiated activities, and provide real-time feedback.

Some key features of the generated pathways were the suggested scientific investigations and the flexible adjustment of the complexity of tasks, offering hints, or proposing simulating virtual experiments when laboratory resources are limited. ChatGPT and Llama are both enabled to assist teachers in adapting IBL instruction. However, ChatGPT tends to outperform Llama in general reasoning, instruction-following, and nuanced responses. ChatGPT 4.0 is more polished for educational material as it currently integrates tools like Microsoft Co-Pilot to generate better and higher-quality adaptive content. Though Llama 3.1 is good, it often requires integration into custom tools or platforms, and the quality can vary based on how it is structured. Hence, the Chat GPT adaptive pathways gave an advanced offering of personalising the content and showcasing IBL.

Overall, these models enhance inclusivity and learner autonomy in science classrooms by offering on-demand, contextualised support throughout the IBL process, thus making science learning more accessible, engaging, and inquiry-driven.

Also, from the detailed analysis of the content, it was clear that ChatGPT4.0 provided rather detailed and more extensive differentiation parameters when compared to Llama 3.1. The highlights from the data show that in creating adaptive IBL learning pathways, the following parameters should always be at the fore for a teacher: clear learning objectives must be set, multiple representations of the content using text, visual, and audio must be employed; Different types of tasks must be considered; There must be an integration of individual and group task; Collaborations should still be promoted even if tasks are differentiated, and the learning objectives must remain the same for all students. What this means for teachers who desire differentiated instruction is that it is the pathway to attaining the objectives and the pace that should be differentiated when considering adaptive and personalised learning pathways, and not the learning objectives and other engagements within learning environments.

The remediation needs of students were used to prompt the GAI chatbots after all the subjects had been catered for, as the last prompt of the six standard prompts. Both GAI engines provided remedial IBL tasks for cases of students who needed remediation. The generated pathways were designed to be dynamic, with the GAI tools adjusting the complexity and focus of inquiry tasks based on the learner's progress and responses.

Teacher views of adaptive learning pathways

Based on Yin (2018), five parts of analysing case study data (pattern matching, explanation building, time-series analysis, logic models, and cross-case synthesis), transcribed textual data from science teachers' interviews were analysed. Six key questions were posed (see Appendix A), and data categorised from the five individual semi-structured interviews conducted with five practising science teachers (Pseudonyms P1-P5) revealed several insights about the GAI-generated adaptive learning pathways that can be summarised in Figure 3 below.

From Figure 3, the six key categories excluding challenges that emanated from the analysed interview data included:

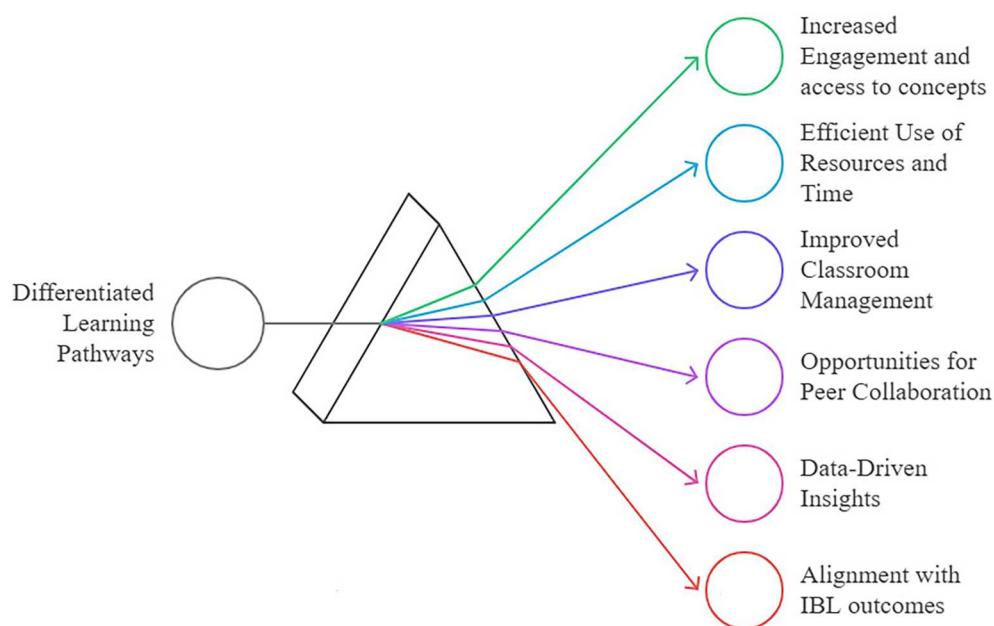


Figure 3. Summarised visual of teachers' evaluation of learning pathways.

Increased engagement and access to content

From the first question, 'How has your experience of using the different GAI tools for creating differentiated learning pathways in supporting student engagement in IBL been? Participant teachers indicated that the adaptive learning pathways that they utilised for their specific subjects enabled them to engage students more during IBL tasks, irrespective of their cognitive abilities or prior knowledge for example, teacher P1 mentioned that *'the pathways enable me to have new ideas on how I could use virtual labs to differentiate instruction for different learners, at different times in the same lesson. This brought more life to the lesson, and I could see that even the quiet learners were engaged'* This excerpt was supported by teacher P3, who indicated that personalised learning activities gave them ideas on how to offer structured support for students with lower levels of prior knowledge of the concepts being taught. This inclusivity observed in the lessons where the pathways were followed was a highlight, especially for teacher P3, whose classroom accommodates students from different socio-economic and cognitive backgrounds.

Efficient use of resources and time

Answering the question on the benefits and challenges throughout the experience, teachers like P2 and P4, who complain of limited resources for traditional face-to-face inquiry-based learning, appreciated the suggestions of including virtual resources like simulations and virtual laboratories. Teacher P4 teaches Physical Sciences (Physics and Chemistry), and he indicated that *'The AI-generated ideas gave me insights on how I could use PhET simulations for my teaching to reinforce my plans for traditional inquiry. I have used PhET simulations before, but after this exercise, I am able to maximise its potential for the learning experiences of different learners in my class, even though we don't have a functional lab at the school'*. P2 added that virtual tools save time that

could be spent going to other schools to gather resources for lab-based inquiry, and it was good to see that GAI also recognised this.

Classroom management and support for different needs

Another benefit teachers realised was that the learning pathways helped with differentiated instruction, which kept all learners involved in one task or the other, hence reducing the distractions that usually arose when learners were bored or overwhelmed. The suggestions from GAI provided guidelines to scaffold ideas for students with low prior knowledge while still catering for those with accurate prior knowledge and cognitive ability with the deployment of more advanced tasks. P2 said: *‘This differentiation of learning tasks was not something I had ever considered due to fear that students might not attain their learning objectives, but for the first time, the learning pathways suggested made me see that segmentation of tasks was an excellent way to keep all learners actively involved, hence reducing disruptions and the need for unproductive noise-making’*. These insights revealed that not only was classroom management maintained within the classroom, but also different student needs were met.

Peer collaboration and leadership

Adding cycles of group work and peer discussions in the adaptive learning pathways, especially those suggested by Llama 3.1, encouraged peer-to-peer support. Teachers did not report any limitations with learner-to-learner collaborations when using differentiated instructional pathways. In fact, teacher P5 indicated that: *‘The task I added for sharing ideas in small groups of learners with mixed cognitive abilities brought out good collaboration skills from all learners, with others just taking charge of the groups as they led conversations around the force of gravity’*. It was clear from participants that creating roles for more knowledgeable students to mentor their peers could rather enhance collaborative learning and develop student autonomy, hence relegating the role of the teacher to the facilitator of learning rather than a transmitter.

Data-informed insights about learning

The use of the suggested formative assessment steps from the adaptive, personalised learning pathways aided teachers in retrieving and using data for each student’s learning milestones. The feedback from the formative assessments created a loop that aided the refinement of the differentiated IBL instruction as lessons were covered further, enabling timely interventions and individualised feedback. Teacher P2 actually said something worth noting about formative assessments with the personalised learning pathways. She said, *‘Because the tasks were varied, for each learning task, I had to check in with learners more often, gathering information on how well each learner is progressing and providing the needed support and individualised feedback. This was something new to me and I enjoyed this approach’*.

Alignment with IBL learning outcomes

Finally, all five teachers indicated that the adaptive learning pathways generated mainly by ChatGPT 4.0 addressed some of the gaps they experienced in attaining learning outcomes and creating tasks for IBL. Indicating that they all love to use IBL but are not always sure of ways to go about it, the pathways suggested activities by ChatGPT 4.0,

especially provided a good instructional model for inquiry that could help learners think critically, solve problems and collaborate with their peers in inquiry tasks. The differentiated pathways, as suggested by ChatGPT 4.0, met some of the fundamental goals of inquiry by providing the relevant scaffold to encourage students to investigate scientific phenomena actively, thereby promoting more engagement with science concepts in line with constructivism and IBL principles.

Challenges of creating personalised adaptive learning pathways

Despite all the benefits identified by science teachers during this research, some challenges were reported in three areas, namely:

Difficulty in keeping track of all the differentiated tasks and assessing learners at the appropriate level suitable for their learning pathway. Teachers indicated that when instruction is differentiated, it becomes automatic that assessments will be differentiated while ensuring that the same learning objectives are met. They found this to be a daunting task, but it was also one that could be mitigated by employing GAI tools further and gaining new knowledge on how to ease the load.

The management of allocated curriculum time in the context of this study also posed a threat to personalised IBL adaptive learning pathways. Teachers highlighted that the annual teaching plan (ATP) is a strict document that does not cater for flexibility in the time allocated for specific subject matter content to be covered. For example, Llama 3.1 suggested that teachers utilise longer periods of time to teach concepts that have only been allocated two weeks in the annual teaching plan. For example, teacher P4 reported, *'I had difficulty aligning the suggested six-week timeline by Llama 3.1, for teaching what the CAPS curriculum and the ATP suggested for forces and Motion, so I had to further adapt the learning pathway and re-prompt to get a more compressed suggestion'*. This poses a limitation to differentiating instruction for students with different learning abilities and contextual backgrounds. Nevertheless, teachers indicated that this challenge could be easily addressed by teachers by refining the adaptive learning pathways to suit their teaching plans.

Lastly, not all the resources from the proposed pathways were available for implementing some of the proposed differentiated tasks. Teachers indicated they were looking forward to how the pathways could cater for improvisation in IBL instruction. This challenge was addressed by one of the researchers, who indicated that further prompting of the bot would always help teachers refine their needs in a way that caters to all scenarios that they could possibly anticipate.

Limitations of the GAI tools when creating adaptive pathways

ChatGPT 4.0 and Llama 3.1 both offer potential for IBL instructional design and adaptation in science education, but their limitations in creating adaptive pathways reveal distinct challenges. ChatGPT 4.0, a multimodal model, excels in generating coherent, contextually relevant responses, supporting personalised learning through real-time feedback and analogies tailored to student needs. However, it struggles with complex, open-ended problem-solving in advanced science topics within Physics, potentially limiting its ability to adapt pathways for diverse learner needs.

Ethical concerns, including biased responses and privacy issues, further complicate its integration into IBL (Ngo, 2023). Llama 3.1, an open-source model, offers cost-efficiency and customisation, making it adaptable for specific science curricula. Yet, it restricts multimodal engagement, crucial for hands-on IBL activities. Some of its suggestions were rather generic and even when further prompted, not much was gathered.

LLM models risk educators becoming over-reliant, potentially reducing critical thinking if not mediated by educators. Integrating these tools requires careful design to ensure adaptive, ethical, and effective IBL pathways.

Discussion and recommendations

The findings from this study show how well LLM-based GAI engines can be used to address the individual learning needs of students in IBL science classrooms, as also suggested by Erduran and Levrini (2024) and seen in a similar English language study by Yeh (2025). All the generated adaptive learning pathways showed great potential in effectively supporting IBL in school science classrooms, though it is important that human oversight is maintained in implementing these learning pathways in science classrooms. The AI-generated learning pathways showed responsiveness to individual learner needs, providing appropriate levels of challenge and scaffolding, as seen in studies like Tapalova and Zhiyenbayeva (2022), to cater for IBL instruction. Practising teachers who verified and implemented the pathways reported higher student engagement and understanding of the scientific concepts explored as they navigated the different learning pathways and the suggested assessment tasks (Smyrnova-Trybulska et al., 2022).

From the different contexts provided, it was clear that LLM-based GAI tools could cater for IBL instruction in different contexts and in cases where learners have varied needs (Cooper, 2023; Smyrnova-Trybulska et al., 2022). Both chatbots could be valuable in integrating diverse pedagogical strategies into the repertoire of science teachers, which is crucial for sustaining the implementation of IBL in school science. The main challenges identified by teachers relate to time constraints, resources for implementation, ethical issues and the accuracy and relevance of some of the GAI-generated biology, chemistry, and physics content in the pathways. All of these setbacks underscore the need for educators to critically evaluate AI-generated outputs to ensure instructional accuracy, as noted by Ferrara (2024).

The teachers also indicated that if accurate student data is captured in the development of the learning pathways, an ethical bridge could arise. This is again congruent with the findings from other studies that propose the careful and ethical use of GAI tools to ensure that data privacy is always maintained (Chen et al., 2020; Ferrara, 2024; Yang, 2022). This finding also fortifies the need for robust frameworks to safeguard student data privacy for the ethical use of GAI tools in education. Hence, we strongly recommend that future implementations incorporate anonymised datasets, as was the case in this study.

Based on the findings of this study, it is recommended that science teachers tap into the power of LLM-based GAI tools to support IBL instruction at the school science level. This could enable students to perform better academically and develop an interest in science careers. Another recommendation is based on the fact that GAI tools will remain machines and are prone to errors. Hence, there is a need for the teacher interface in AI integration to mitigate subject matter content errors and eliminate ethical

violations as far as the data privacy of students is concerned. We also recommend that teacher educators develop professional development programmes to equip teachers with skills to critically evaluate and adapt AI-generated adaptive pathways, ensuring alignment with curricular goals and student needs (Yang, 2022).

Implications for practice

The findings of this study have practical implications for the future of science education, specifically IBL instruction. It is clear from the findings that teachers could enhance instructional efficiency and differentiation by creating personalised learning experiences crafted using GAI tools and the power of LLM. Teachers could also create adaptive learning pathways that make IBL more practical and relatable for students and themselves to elevate the educational experience. However, AI integration in education for content generation must be approached carefully to mitigate ethical and practical challenges.

Also eminent is the need for teachers to constantly adapt instructional practices to future-proof themselves should there be an urgent need to respond to unexpected challenges, such as the sudden transition to remote learning during the COVID-19 pandemic. From the pandemic experience, it was noted that teachers who could pivot quickly to online platforms and modify their instructional strategies were more successful in maintaining continuity in student learning (Trust & Whalen, 2021).

Lastly, incorporating flexibility into their teaching practice also allows educators to create more inclusive classrooms to tailor instruction for diverse learning styles and needs, thereby promoting equity and accessibility in education (Tomlinson, 2014). The ability to adapt not only enhances teaching effectiveness but also contributes to the resilience and overall success of both teachers and students in a constantly changing educational environment.

Theoretical implications

Vygotsky's (1987) concept of the Zone of Proximal Development (ZPD), combined with the theory of differentiated instruction (Tomlinson, 1999), in this study highlights the range of tasks that learners can perform with the help of a more knowledgeable other, particularly when these tasks are tailored to their individualised learning needs. AI systems can serve as this 'more knowledgeable other' by offering scaffolding to students, but, more interestingly, differentiating these to meet their needs. Adaptive learning technologies powered by GAI can identify the ZPD for each student and provide appropriate support, gradually reducing assistance as the learner gains competence. This aligns with the goal of Personalised learning, ensuring that students are consistently challenged yet supported.

Limitations

The study was conducted in a particular context that caters to a unique curriculum. Therefore, the adaptive pathways generated by GAI tools for teachers may also be varied when different scenarios and contexts are considered. The small sample size ($n = 5$) used in this case study limits the generalisability of the findings of the study.

Future research with larger, more diverse teacher cohorts is recommended. Only two GAI tools were analysed, creating a limitation on how the findings could be generalised for other GAI tools. A quantitative approach (analysis of student performance/achievement data) to indirectly analyse the effectiveness of GAI tools in creating adaptive learning pathways for students was not attempted, but should be considered for future studies.

Conclusion

This research highlights the potential of generative AI LLMs to affect inquiry-based learning pedagogy by creating adaptive learning pathways. This study shows that GAI tools like Chat GPT4.0 and Llama 3.1 could enhance IBL pedagogy by supporting teachers to create personalised, equitable learning environments. However, their successful integration hinges on ethical safeguards and teacher training. While the use of AI in education presents new opportunities for personalised learning, it also necessitates carefully considering the limitations and ethical implications. A collaborative approach, which includes all stakeholders (teachers, technologists and policy makers), is crucial for AI integration while ensuring the integrity of educational practices. Further research is needed to refine GAI models, enhance accuracy, and explore their long-term impact on student learning and teacher practice.

Disclosure statement

No potential conflict of interest was reported by the author(s).

Ethics statement

This research follows the ethical principles and guidelines for academic integrity in research as benchmarked internationally and at the university in which the study was conducted. Participants were fully informed about the purpose of the study, procedures, and potential risks, and their informed consent was obtained. Confidentiality and anonymity of participant data were maintained throughout the research process. The study received approval from the faculty of education research ethics committee (Anonymised) under clearance number SEM 1-2024-125. The researchers ensured that bias was minimised and objectivity maintained in the data collection and analysis. No conflicts of interest or external influences impacted the findings of the study. This research aligns with principles of respect, beneficence, and justice, prioritising participant well-being and academic rigour.

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References

- Akuma, F. V., & Callaghan, R. (2019). A systematic review characterizing and clarifying intrinsic teaching challenges linked to inquiry-based practical work. *Journal of Research in Science Teaching*, 56(5), 619–648. <https://doi.org/10.29333/ejmste/11352>

- Al-Hamdani, D., & Yousif, J. H. (2025). Artificial intelligence revolution for enhancing modern education using zone of proximal development approach. *Applied Computing Journal*, 5(2), 386–398. <https://doi.org/10.52098/acj.20255239>
- Author. (2016).
- Bernacki, M. L., Greene, M. J., & Lobczowski, N. G. (2021). A systematic review of research on personalized learning: Personalized by whom, to what, how, and for what purpose (s)? *Educational Psychology Review*, 33(4), 1675–1715. <https://doi.org/10.1007/s10648-021-09615-8>
- Brown, T. B. (2020). Language models are few-shot learners. *arXiv preprint ArXiv:2005.14165*. <https://doi.org/10.48550/arXiv.2005.14165>
- Chen, X., Xie, H., Zou, D., & Hwang, G.-J. (2020). Application and theory gaps during the rise of artificial intelligence in education. *Computers and Education: Artificial Intelligence*, 1, 100002. <https://doi.org/10.1016/j.caeai.2020.100002>
- Cohen, J. (1960). A coefficient of agreement for nominal scales. *Educational and Psychological Measurement*, 20(1), 37–46. <https://doi.org/10.1177/001316446002000104>
- Cooper, G. (2023). Examining science education in ChatGPT: An exploratory study of generative artificial intelligence. *Journal of Science Education and Technology*, 32(3), 444–452. <https://doi.org/10.1007/s10956-023-10039-y>
- Darling-Hammond, L., Barron, B., Pearson, P. D., Schoenfeld, A. H., Stage, E. K., Zimmerman, T. D., Cervetti, G. N., & Tilson, J. L. (2008). *Powerful learning: What we know about teaching for understanding*. Jossey-Bass.
- Erduran, S., & Levirini, O. (2024). The impact of artificial intelligence on scientific practices: An emergent area of research for science education. *International Journal of Science Education*, 46(18), 1982–1989. <https://doi.org/10.1080/09500693.2024.2306604>
- Ferrara, E. (2024). GenAI against humanity: Nefarious applications of generative artificial intelligence and large language models. *Journal of Computational Social Science*, 7(1), 549–569. <https://doi.org/10.1007/s42001-024-00250-1>
- Forbes, C. T., Neumann, K., & Schiepe-Tiska, A. (2020). Patterns of inquiry-based science instruction and student science achievement in PISA 2015. *International Journal of Science Education*, 42(5), 783–806. <https://doi.org/10.1080/09500693.2020.1730017>
- Gholam, A. P. (2019). Inquiry-based learning: Student teachers' challenges and perceptions. *Journal of Inquiry and Action in Education*, 10(2), 112–133.
- Gligorea, I., Cioca, M., Oancea, R., Gorski, A. T., Gorski, H., & Tudorache, P. (2023). Adaptive learning using artificial intelligence in e-learning: A literature review. *Education Sciences*, 13(12), 1216. <https://doi.org/10.3390/educsci13121216>
- Gupta, P., Ding, B., Guan, C., & Ding, D. (2024). Generative AI: A systematic review using topic modelling techniques. *Data and Information Management*, 8(2), 100066. <https://doi.org/10.1016/j.dim.2024.100066>
- Hmelo-Silver, C. E., Duncan, R. G., & Chinn, C. A. (2007). Scaffolding and achievement in problem-based and inquiry learning: A response to Kirschner, Sweller, and Clark. *Educational Psychologist*, 42(2), 99–107. <https://doi.org/10.1080/00461520701263368>
- Landis, J. R., & Koch, G. G. (1977). The measurement of observer agreement for categorical data. *Biometrics*, 159–174. <https://doi.org/10.2307/2529310>
- Lazonder, A. W., & Harmsen, R. (2016). Meta-analysis of inquiry-based learning: Effects of guidance. *Review of Educational Research*, 86(3), 681–718. <https://doi.org/10.3102/0034654315627366>
- Lederman, J. S., Lederman, N. G., Bartels, S., Jimenez, J., Acosta, K., Akubo, M., Aly, S., Andrade, M. A. B. S. D., Atanasova, M., Blanquet, E., Blonder, R., Brown, P., Cardoso, R., Castillo-Urueta, P., Chaipidech, P., Concannon, J., Dogan, O. K., El-Deghaidy, H., Elzorkani, A., ... Wishart, J. (2021). International collaborative follow-up investigation of graduating high school students' understandings of the nature of scientific inquiry: Is progress being made? *International Journal of Science Education*, 43(7), 991–1016. <https://doi.org/10.1080/09500693.2021.1894500>
- Loertscher, D. V., & Young, M. (2025). Models of inquiry and the learner: A fresh look. *Learning Hub*, 1(2), 2.

- Mesci, G., Çavuş-Güngören, S., & Yesildag-Hasancebi, F. (2020). Investigating the development of pre-service science teachers' NOSI views and related teaching practices. *International Journal of Science Education*, 42(1), 50–69. <https://doi.org/10.1080/09500693.2019.1700316>
- Meta. (2024). <https://www.meta.ai/>
- Minner, D. D., Levy, A. J., & Century, J. (2010). Inquiry-based science instruction—What is it and does it matter? Results from a research synthesis years 1984 to 2002. *Journal of Research in Science Teaching*, 47(4), 474–496. <https://doi.org/10.1002/tea.20347>
- National Research Council. (2012). *A framework for K-12 science education: Practices crosscutting concepts and core ideas*. National Academy Press.
- Ngo, T. T. A. (2023). The perception by university students of the use of ChatGPT in education. *International Journal of Emerging Technologies in Learning*, 18(17), 4. <https://doi.org/10.3991/ijet.v18i17.39019>
- OpenAI. (2024). ChatGPT. <https://openai.com/chatgpt/overview/>
- Radford, A., Wu, J., Child, R., Luan, D., Amodei, D., & Sutskever, I. (2019). Language models are unsupervised multitask learners. *OpenAI blog*, 1(8), 9. <https://storage.prod.researchhub.com/uploads/papers/2020/06/01/language-models.pdf>
- Rojas, M. P., & Chiappe, A. (2024). Artificial intelligence and digital ecosystems in education: A review. *Technology, Knowledge and Learning*, 29(4), 2153–2170. <https://doi.org/10.1007/s10758-024-09732-7>
- Smagorinsky, P. (2018). Deconflating the ZPD and instructional scaffolding: Retranslating and reconceiving the zone of proximal development as the zone of next development. *Learning, culture and social interaction*, 16, 70–75. <https://doi.org/10.1016/j.lcsi.2017.10.009>
- Smyrnova-Trybulska, E., Morze, N., & Varchenko-Trotsenko, L. (2022). Adaptive learning in university students' opinions: Cross-border research. *Education and Information Technologies*, 27(5), 6787–6818. <https://doi.org/10.1080/09500693.2019.1700316>
- Tang, H., Xing, W., & Pei, B. (2019). Time really matters: Understanding the temporal dimension of online learning using educational data mining. *Journal of Educational Computing Research*, 57(5), 1326–1347. <https://doi.org/10.1177/0735633118784705>
- Tang, K. S. (2024). Informing research on generative artificial intelligence from a language and literacy perspective: A meta-synthesis of studies in science education. *Science Education*, 108(5), 1329–1355. <https://doi.org/10.1002/sce.21875>
- Tapalova, O., & Zhiyenbayeva, N. (2022). Artificial intelligence in education: AIED for personalised learning pathways. *Electronic Journal of e-Learning*, 20(5), 639–653. <https://doi.org/10.34190/ejel.20.5.2597>
- Tomlinson, C. (1999). *Differentiated classroom: Responding to the needs of all learners*. Association for Supervision and Curriculum Development. <https://cmc.marmot.org/Record/b19383150>
- Tomlinson, C. (2001). *How to differentiate instruction in mixed-ability classrooms* (2nd ed.). Association for Supervision and Curriculum Development.
- Tomlinson, C. A. (2014). *The differentiated classroom: Responding to the needs of all learners*. Ascd.
- Trust, T., & Whalen, J. (2021). Emergency remote teaching with technology during the COVID-19 pandemic: Using the whole teacher lens to examine educator's experiences and insights. *Educational Media International*, 58(2), 145–160. <https://doi.org/10.1080/09523987.2021.1930479>
- Usher, M., & Barak, M. (2024). Unpacking the role of AI ethics online education for science and engineering students. *International Journal of STEM Education*, 11(1), 35. <https://doi.org/10.1186/s40594-024-00493-4>
- Vygotsky, L. S. (1987). *The collected works of LS Vygotsky: Problems of the theory and history of psychology* (Vol. 3). Springer Science & Business Media.
- Walkington, C., & Bernacki, M. L. (2020). Appraising research on personalized learning: Definitions, theoretical alignment, advancements, and future directions. *Journal of Research on Technology in Education*, 52(3), 235–252. <https://doi.org/10.1080/15391523.2020.1747757>
- Wen, Q., & Cai, J. (2024). Applying structural equation modeling to examine the role of teacher beliefs and practices in differentiated instruction in physical education: Multiple mediation analyses. *Psychology in the Schools*, 61(7), 3045–3062. <https://doi.org/10.1002/pits.23206>

- Xu, W., & Ouyang, F. (2022). A systematic review of AI role in the educational system based on a proposed conceptual framework. *Education and Information Technologies*, 27(3), 4195–4223. <https://doi.org/10.1007/s10639-021-10774-y>
- Yang, M., & Weng, F. (2023). AI-Powered Personalized learning journeys: Revolutionizing information management for college students in online platforms. *Journal of Information Systems Engineering and Management*, 8(1), 23196. <https://doi.org/10.55267/iadt.07.14079>
- Yang, W. (2022). Artificial intelligence education for young children: Why, what, and how in curriculum design and implementation. *Computers and Education: Artificial Intelligence*, 3, 100061. <https://doi.org/10.1016/j.caeai.2022.100061>
- Yeh, H. C. (2025). The synergy of generative AI and inquiry-based learning: Transforming the landscape of English teaching and learning. *Interactive Learning Environments*, 33(1), 88–102. <https://doi.org/10.1080/10494820.2024.2335491>
- Yeoh, C. P., Li, C. T., & Hou, H. T. (2025). Game-based collaborative scientific inquiry learning using realistic context and inquiry process-based multidimensional scaffolding. *International Journal of Science Education*, 47(8), 961–983. <https://doi.org/10.1080/09500693.2024.2354944>
- Yin, R. K. (2018). *Case study research and applications: Design and methods* (6th ed.). SAGE Publications.
- Zhai, X. (2023). ChatGPT for next generation science learning. *XRDS: Crossroads, The ACM Magazine for Students*, 29(3), 42–46. <https://doi.org/10.1145/3589649>
- Zhai, X., & Nehm, R. (2023). AI and formative assessment: The train has left the station. *Journal of Research in Science Teaching*, 60(6), 1390–1398. <https://doi.org/10.1002/tea.21885>
- Zhang, X., Sun, J., & Deng, Y. (2023). Design and application of intelligent classroom for English language and literature based on artificial intelligence technology. *Applied Artificial Intelligence*, 37(1), 2216051. <https://doi.org/10.1080/08839514.2023.2216051>

Appendix A: Semi-structured interview questions

1. How has your experience of using the different GAI tools for creating differentiated learning pathways in supporting student engagement in IBL been?
2. Were there any benefits or Challenges throughout the experience?
3. Collaborative learning is a key aspect of IBL instruction. Was this aspect affected in any way since the goal was to personalise instruction? Explain.
4. What was the role of the GAI-generated pathways in suggesting adequate formative assessments and feedback?
5. Looking at your own practice, which of the GAI tools would help you implement differentiated IBL outcomes more effectively in your classroom?
6. In conclusion, could you provide any further insights on how teachers can use GAI to support inquiry instruction based on your experience so far?